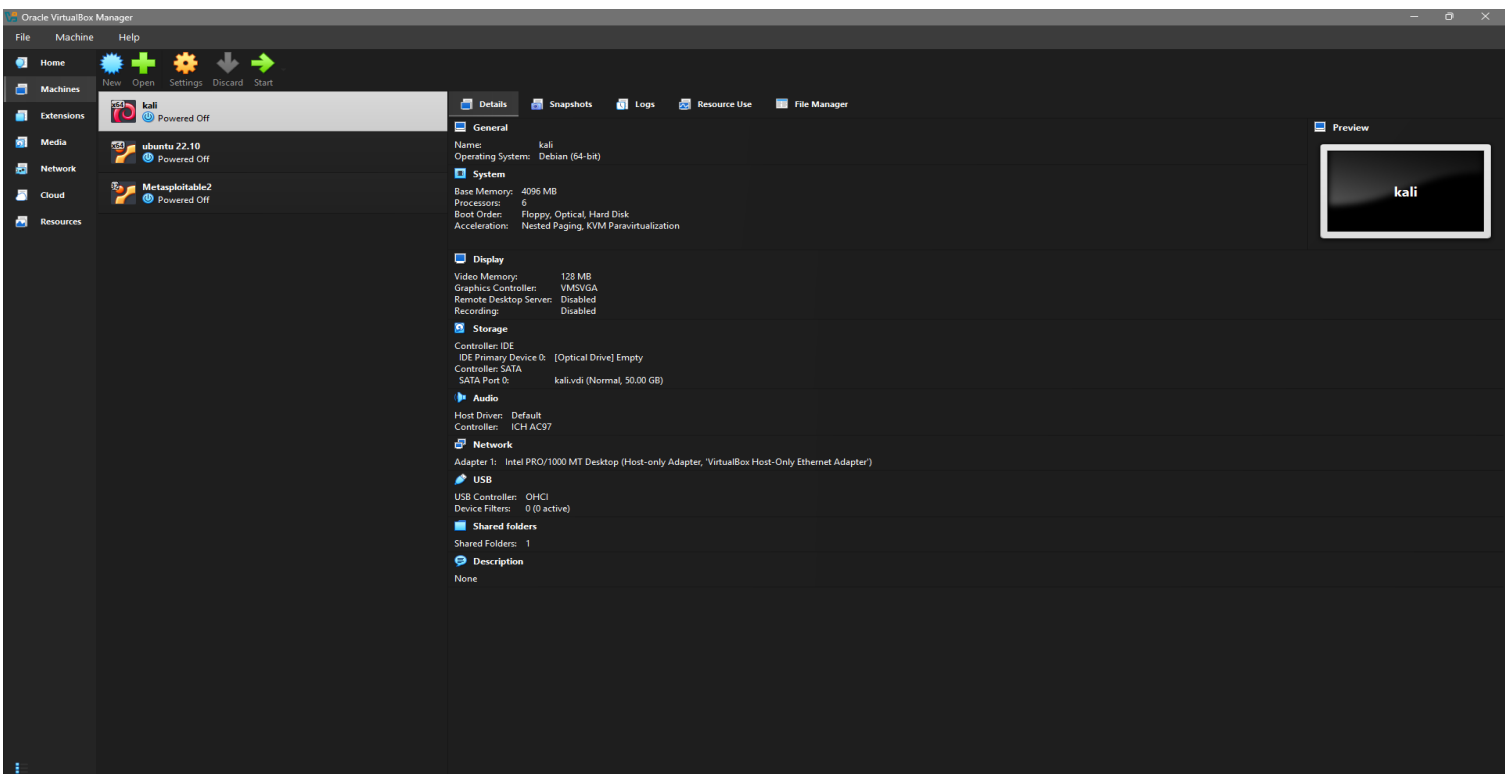
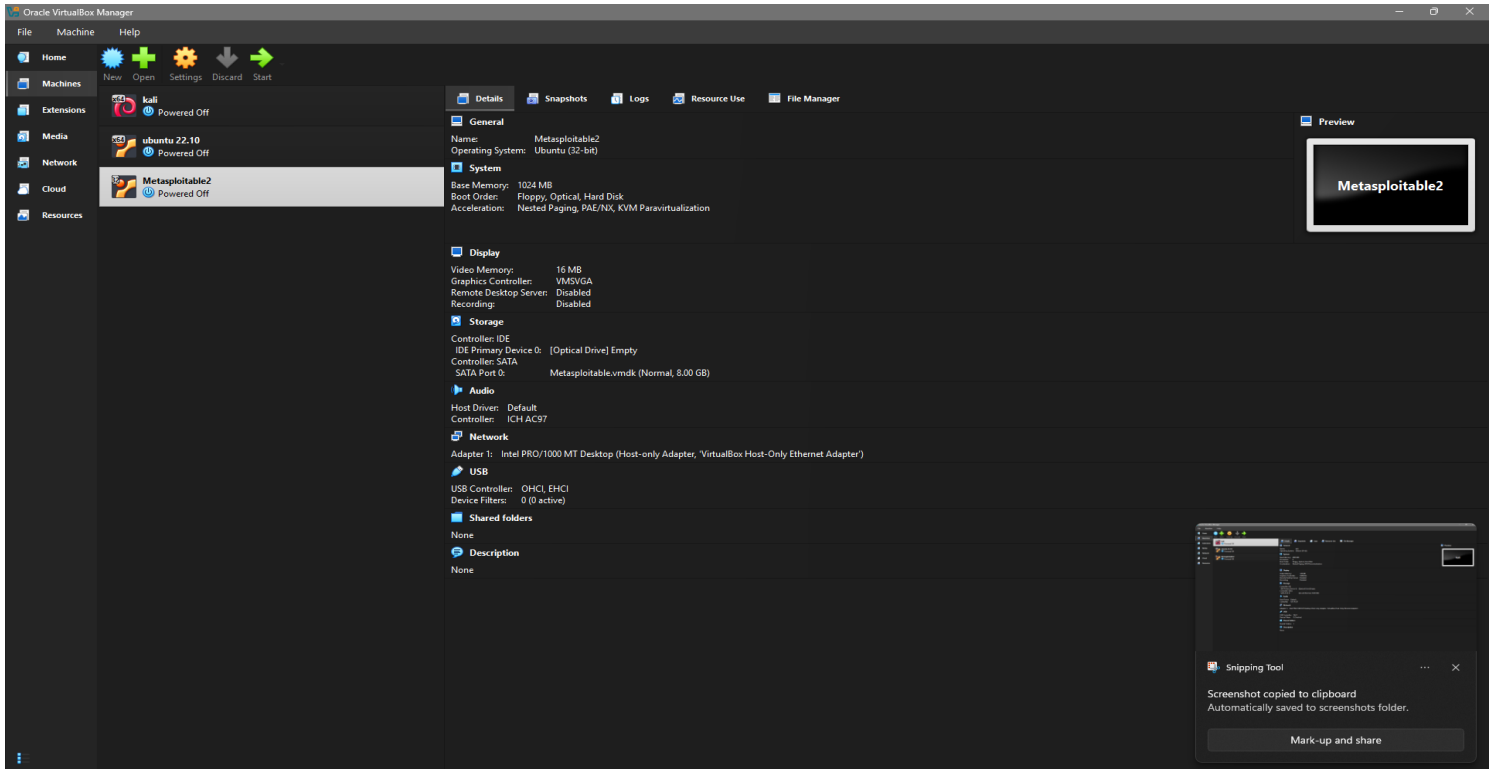


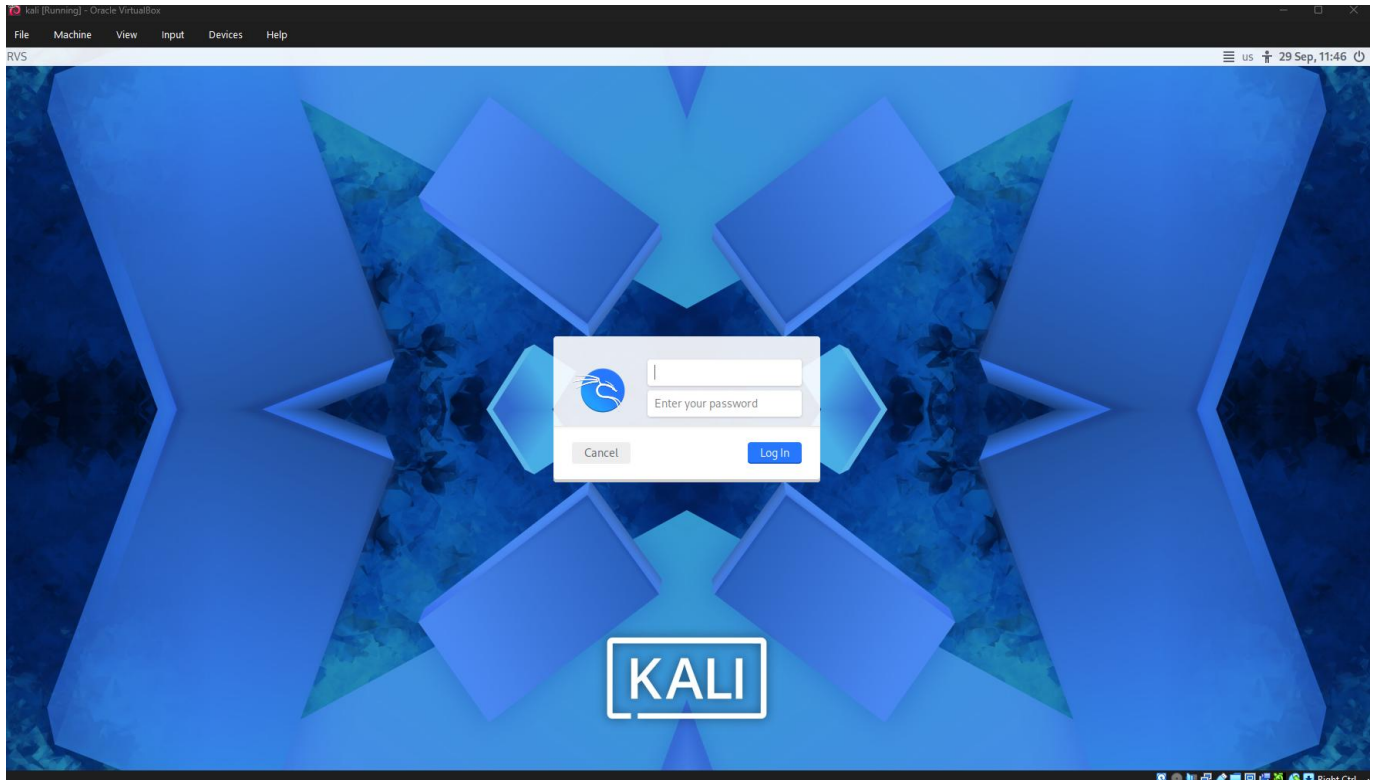
Task 1 – Practical Screenshots

Foundation & Environment Setup

1. VirtualBox Metasploitable2 and Kali Network Configur



2. Metasploitable2 and Kali Login



```
Metasploitable2 [Running] - Oracle VM VirtualBox

Contact: msfdev[at]metasploit.com

Login with msfadmin/msfadmin to get started

metasploitable login: msfadmin
Password:
Last login: Tue Sep 29 00:50:33 EDT 2026 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

msfadmin@metasploitable:~$
```

3. Kali to Metasploitable2: Ping, Traceroute and Netstat

```
kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4

rvs@RVS: ~
Session Actions Edit View Help

(rvs@RVS)~$ ping -c 3 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data.
64 bytes from 192.168.56.102: icmp_seq=1 ttl=64 time=2.55 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=64 time=1.89 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=64 time=0.851 ms

--- 192.168.56.102 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2044ms
rtt min/avg/max/mdev = 0.851/1.763/2.545/0.697 ms

(rvs@RVS)~$ traceroute 192.168.56.102
traceroute to 192.168.56.102 (192.168.56.102), 30 hops max, 60 byte packets
 1 192.168.56.102 (192.168.56.102) 2.058 ms 1.963 ms 1.932 ms

(rvs@RVS)~$ netstat -an | grep 192.168.56.102

(rvs@RVS)~$ netstat -an
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
udp        0      0 192.168.56.101:68      192.168.56.100:67      ESTABLISHED
raw6       0      0 :::58                  :::*                     7

Active UNIX domain sockets (servers and established)
Proto RefCnt Flags               Type               State         I-Node   Path
unix 3      [ ]          STREAM            CONNECTED      11767
unix 3      [ ]          STREAM            CONNECTED      44281    /run/systemd/journal/stdout
unix 3      [ ]          STREAM            CONNECTED      17491
unix 3      [ ]          STREAM            CONNECTED      13499
unix 3      [ ]          STREAM            CONNECTED      13302    /run/systemd/journal/stdout
unix 3      [ ]          STREAM            CONNECTED      11763
unix 3      [ ]          STREAM            CONNECTED      16978
unix 3      [ ]          STREAM            CONNECTED      15615
unix 3      [ ]          STREAM            CONNECTED      12660
unix 3      [ ]          STREAM            CONNECTED      12084    /run/user/1000/bus
unix 3      [ ]          STREAM            CONNECTED      13907
unix 3      [ ]          STREAM            CONNECTED      15529
unix 3      [ ]          STREAM            CONNECTED      16151
unix 3      [ ]          STREAM            CONNECTED      11589    /run/user/1000/pipewire-0-manager
unix 3      [ ]          STREAM            CONNECTED      5792
unix 3      [ ]          STREAM            CONNECTED      12014
unix 3      [ ]          STREAM            CONNECTED      14183
unix 3      [ ]          STREAM            CONNECTED      14463    /run/user/1000/bus
unix 3      [ ]          STREAM            CONNECTED      13841    /run/user/1000/bus
unix 3      [ ]          STREAM            CONNECTED      11478    /run/dbus/system_bus_socket
unix 3      [ ]          STREAM            CONNECTED      13294    /run/systemd/journal/stdout
```

4. Linux File System Navigation: ls, cd and pwd

```
kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4

rvs@RVS: ~
Session Actions Edit View Help

(rvs@RVS)~$ ls
Chankro  decrypt.txt  Desktop  Documents  Downloads  encrypted.txt  gif.txt  hash.txt  index.html  Music  Pictures  Projects  Public  pwnds.zip  Templates  text.txt  tryhackme.gif.php  Videos

(rvs@RVS)~$ cd Chankro

(rvs@RVS)~/Chankro$ pwd
/home/rvs/Chankro

(rvs@RVS)~/Chankro$ ls -l
total 84
-rw-rw-r-- 1 rvs rvs 2436 Aug 18 16:01 chankro.py
-rw-rw-r-- 1 rvs rvs  0 Aug 19 14:51 c.sh
-rw-rw-r-- 1 rvs rvs  8 Aug 19 15:21 gif.txt
-rw-rw-r-- 1 rvs rvs 7292 Aug 18 16:01 hook32.so
-rw-rw-r-- 1 rvs rvs 8504 Aug 18 16:01 hook64.so
-rw-rw-r-- 1 rvs rvs  444 Aug 18 16:01 hook.c
-rw-rw-r-- 1 rvs rvs 35141 Aug 18 16:01 LICENSE
-rw-rw-r-- 1 rvs rvs  904 Aug 18 16:01 README.md
-rw-rw-r-- 1 rvs rvs 11674 Aug 19 15:30 tryhackme.gif.php

(rvs@RVS)~/Chankro$ touch rvs.txt

(rvs@RVS)~/Chankro$ nano rvs.txt

(rvs@RVS)~/Chankro$ cat rvs.txt
hi i am vikash kumar blah blah

(rvs@RVS)~/Chankro$ cd ../

(rvs@RVS)~$ pwd
/home/rvs
```

5. File Ownership and Permissions: chown and chmod

```
kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
rvs@RVS: ~/Chankro

Session Actions Edit View Help

(rvs@RVS)-[/home]
$ pwd
/home

(rvs@RVS)-[/home]
$ ls
rh rh1 rvs student3

(rvs@RVS)-[/home]
$ cd ~

(rvs@RVS)-[~]
$ ls
Chankro decrypt.txt Desktop Documents Downloads encrypted.txt gif.txt hash.txt index.html Music Pictures Projects Public pwds.zip Templates text.txt tryhackme.gif.php Videos

(rvs@RVS)-[~]
$ sudo chown rh1 rvs.txt
chown: cannot access 'rvs.txt': No such file or directory

(rvs@RVS)-[~]
$ cd Chankro

(rvs@RVS)-[~/Chankro]
$ ls
chankro.py c.sh gif.txt hook32.so hook64.so hook.c LICENSE README.md rvs.txt tryhackme.gif.php

(rvs@RVS)-[~/Chankro]
$ sudo chown rh1 rvs.txt

(rvs@RVS)-[~/Chankro]
$ ls -l
total 88
-rw-rw-r-- 1 rvs rvs 2436 Aug 18 16:01 chankro.py
-rw-rw-r-- 1 rvs rvs 0 Aug 19 14:51 c.sh
-rw-rw-r-- 1 rvs rvs 8 Aug 19 15:21 gif.txt
-rw-rw-r-- 1 rvs rvs 7292 Aug 18 16:01 hook32.so
-rw-rw-r-- 1 rvs rvs 8504 Aug 18 16:01 hook64.so
-rw-rw-r-- 1 rvs rvs 444 Aug 18 16:01 hook.c
-rw-rw-r-- 1 rvs rvs 35141 Aug 18 16:01 LICENSE
-rw-rw-r-- 1 rvs rvs 904 Aug 18 16:01 README.md
-rw-rw-r-- 1 rh1 rvs 31 Sep 29 11:20 rvs.txt
-rw-rw-r-- 1 rvs rvs 11674 Aug 19 15:30 tryhackme.gif.php
```

```
kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
root@RVS: /home/rvs/Chankro

Session Actions Edit View Help

(rvs@RVS)-[~/Chankro]
$ sudo su
(root@RVS)-[/home/rvs/Chankro]
$ ls
chankro.py c.sh gif.txt hook32.so hook64.so hook.c LICENSE README.md rvs.txt tryhackme.gif.php

(root@RVS)-[/home/rvs/Chankro]
$ ls -l
total 88
-rw-rw-r-- 1 rvs rvs 2436 Aug 18 16:01 chankro.py
-rw-rw-r-- 1 rvs rvs 0 Aug 19 14:51 c.sh
-rw-rw-r-- 1 rvs rvs 8 Aug 19 15:21 gif.txt
-rw-rw-r-- 1 rvs rvs 7292 Aug 18 16:01 hook32.so
-rw-rw-r-- 1 rvs rvs 8504 Aug 18 16:01 hook64.so
-rw-rw-r-- 1 rvs rvs 444 Aug 18 16:01 hook.c
-rw-rw-r-- 1 rvs rvs 35141 Aug 18 16:01 LICENSE
-rw-rw-r-- 1 rvs rvs 904 Aug 18 16:01 README.md
-rw-rw-r-- 1 rh1 rvs 31 Sep 29 11:20 rvs.txt
-rw-rw-r-- 1 rvs rvs 11674 Aug 19 15:30 tryhackme.gif.php

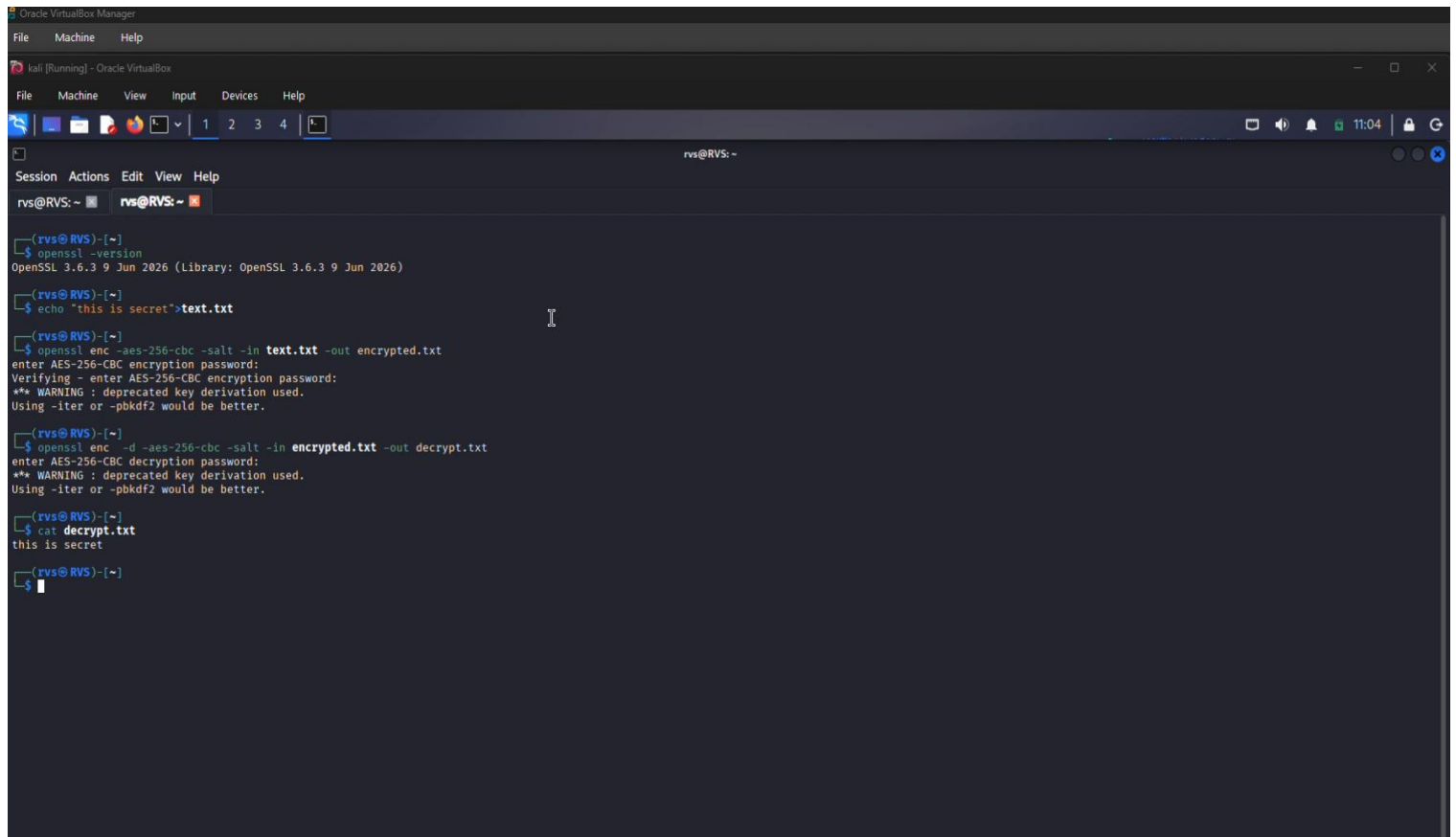
(root@RVS)-[/home/rvs/Chankro]
$ chmod u-w rvs.txt

(root@RVS)-[/home/rvs/Chankro]
$ ls -l
total 88
-rw-rw-r-- 1 rvs rvs 2436 Aug 18 16:01 chankro.py
-rw-rw-r-- 1 rvs rvs 0 Aug 19 14:51 c.sh
-rw-rw-r-- 1 rvs rvs 8 Aug 19 15:21 gif.txt
-rw-rw-r-- 1 rvs rvs 7292 Aug 18 16:01 hook32.so
-rw-rw-r-- 1 rvs rvs 8504 Aug 18 16:01 hook64.so
-rw-rw-r-- 1 rvs rvs 444 Aug 18 16:01 hook.c
-rw-rw-r-- 1 rvs rvs 35141 Aug 18 16:01 LICENSE
-rw-rw-r-- 1 rvs rvs 904 Aug 18 16:01 README.md
-r--r--r-- 1 rh1 rvs 31 Sep 29 11:20 rvs.txt
-rw-rw-r-- 1 rvs rvs 11674 Aug 19 15:30 tryhackme.gif.php

(root@RVS)-[/home/rvs/Chankro]
$ chmod u+w rvs.txt

(root@RVS)-[/home/rvs/Chankro]
$ ls -l
total 88
-rw-rw-r-- 1 rvs rvs 2436 Aug 18 16:01 chankro.py
-rw-rw-r-- 1 rvs rvs 0 Aug 19 14:51 c.sh
-rw-rw-r-- 1 rvs rvs 8 Aug 19 15:21 gif.txt
-rw-rw-r-- 1 rvs rvs 7292 Aug 18 16:01 hook32.so
-rw-rw-r-- 1 rvs rvs 8504 Aug 18 16:01 hook64.so
-rw-rw-r-- 1 rvs rvs 444 Aug 18 16:01 hook.c
-rw-rw-r-- 1 rvs rvs 35141 Aug 18 16:01 LICENSE
-rw-rw-r-- 1 rvs rvs 904 Aug 18 16:01 README.md
```

6. openssl encryption and decryption



The screenshot shows a Kali Linux virtual machine running in Oracle VM VirtualBox Manager. The terminal window displays the following commands and output:

```
(rws@RVS)-[~]  
$ openssl -version  
OpenSSL 3.6.3 9 Jun 2026 (Library: OpenSSL 3.6.3 9 Jun 2026)  
  
(rws@RVS)-[~]  
$ echo "this is secret">text.txt  
  
(rws@RVS)-[~]  
$ openssl enc -aes-256-cbc -salt -in text.txt -out encrypted.txt  
enter AES-256-CBC encryption password:  
Verifying - enter AES-256-CBC encryption password:  
** WARNING : deprecated key derivation used.  
Using -iter or -pbkdf2 would be better.  
  
(rws@RVS)-[~]  
$ openssl enc -d -aes-256-cbc -salt -in encrypted.txt -out decrypt.txt  
enter AES-256-CBC decryption password:  
** WARNING : deprecated key derivation used.  
Using -iter or -pbkdf2 would be better.  
  
(rws@RVS)-[~]  
$ cat decrypt.txt  
this is secret  
  
(rws@RVS)-[~]  
$
```

